

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0557

Ridgely 100 kV
May 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0557. The Interconnection Customer proposes a 363.5 MW natural gas combined cycle facility interconnecting at the Ridgely 100 kV substation in the Ohio area. The request was received on December 22, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 254 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,496,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RIDGELY / ROSEDALE 100kV	RIDGELY / SULIFF 100kV	314.0	112	100.0
SUTLIFF / RIDGELY 100kV L	RIDGELY / ROSEDALE 100kV	289.5	121	100.0
CONGRESS / SUTLIFF 100kV	RIDGELY / ROSEDALE 100kV	267.3	127	100.0
CONGRESS / BRADY 100kV L	RIDGELY / ROSEDALE 100kV	126.7	143	59.2
ROSEDALE SUBSTATION / BILLINGS EASTSIDE 100kV	RIDGELY / SUTLIFF SUBSTATION 100kV L	124.2	111	43.7
GILCHRIST SUBSTATION / CONGRESS 100kV	RIDGELY / ROSEDALE 100kV	117.7	109	40.8
DAVID SWINFORD / TALLMADGE 100kV	BRADY / CONGRESS 100kV	113.5	158	11.4
KNOX / AES 230kV	NORTH CAMBRIDGE / ATASCOCITA 230kV L	112.1	359	5.2
BLACKROCK / CALDWELL 345kV	CALDWELL / CABON CTR SUBSTATION 345kV	104.2	996	5.4
ROSEDALE / MCDONNELL 100kV	SUTLIFF / RIDGELY 100kV line	103.8	99	17.2
SHAWVILLE STA / BLACKROCK SUBSTATION 345kV	CARBON CETER 500 kV/345 kV TX	101.1	1,694	7.4
P.T.S / BURWELL 345kV	ALLEGHENY ENERGY NITS 3 4 & 5 345/1kV Auto	100.7	1,246	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RIDGELY / ROSEDALE 100kV	Rebuild line with higher-capacity conductor	\$3,063,000	100% (\$3,063,000)
SUTLIFF / RIDGELY 100kV L	Rebuild line with higher-capacity conductor	\$1,624,000	100% (\$1,624,000)
CONGRESS / SUTLIFF 100kV	Rebuild line with higher-capacity conductor	\$4,395,000	100% (\$4,395,000)
CONGRESS / BRADY 100kV line	Rebuild line with higher-capacity conductor	\$9,946,000	100% (\$9,946,000)
ROSEDALE / BILLINGS EASTSIDE 100kV line	Rebuild line with higher-capacity conductor	\$1,320,000	100% (\$1,320,000)
CONGRESS / GILCHRIST 100kV	Rebuild line with higher-capacity conductor	\$5,734,000	100% (\$5,734,000)
DAVID SWINFORD / TALLMADGE 100kV	Rebuild line with higher-capacity conductor	\$820,000	88% (\$726,000)
KNOX / AES 230kV	Rebuild line with higher-capacity conductor	\$21,128,000	76% (\$15,964,000)
BLACKROCK / CALDWELL 345kV	Reconductor line	\$19,088,000	18% (\$3,529,000)
MCDONNELL / ROSEDALE 100kV	Reconductor line	\$3,069,000	100% (\$3,069,000)
SHAWVILLE STA / BLACKROCK SUBSTATION 345kV	Reconductor line	\$17,414,000	6% (\$976,000)
P.T.S / BURWELL 345kV	Reconductor line	\$4,284,000	4% (\$150,000)
	Total allocated to project		\$50,496,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.